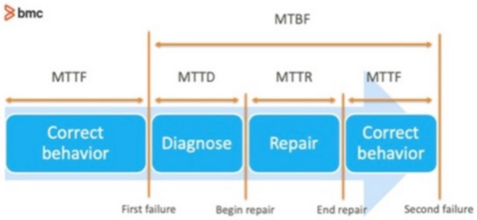
$$MTBF = MTTF + MTTR \qquad MTBF = \int R(t)\,dt$$

$$MTBF = \frac{1}{\lambda}. \qquad R(t) = e^{-\lambda t}$$

$$A = \frac{MTTF}{MTTF + MTTR}$$



## Fiabilitatea r din n

- probabilitatea să funcționeze exact r blocuri :

$$P_n(r) = C_n^r R^r (1-R)^{n-r}$$

- fiabilitatea structurii :

$$R = \sum_{k=r}^{n} C_n^k R^k (1-R)^{n-k}$$

- Rata defecțiunilor

$$h(t) = \frac{1}{MTBF} = \frac{1}{MTTF}$$

- P(A) condiționat de evenimentul B

$$P(A|B) = \frac{P(A\cdot B)}{P(B)}$$

$$f(t) = -\frac{dR(t)}{dt} \quad ; \quad F(t) = \int_0^t f(z)\,dz$$

- Steady - State

dacă x admite f(x) $\Rightarrow E[x] = \int_{-\infty}^{\infty} x f(x)\,dx$

- **Structuri serie**
  - $R_s(t) = \prod_{i=1}^{n} R_i(t)$
  - $\lambda_s = \sum_{i=1}^{n} \lambda_i$
  - $Q_s = 1 - R_s(t) = 1 - \prod_{i=1}^{n} (1 - Q_i(t))$

- **Structuri paralel**
  - $R_p = 1 - \prod_{i=1}^{n} (1 - R_i) = 1 - (1 - R_1)(1 - R_2)\dots(1 - R_n)$
  - $MTBF_p = \frac{1}{\lambda}$

  → la module identice

  - $MTBF_p = \frac{1}{\lambda} \sum_{i=1}^{n} \frac{1}{i}$

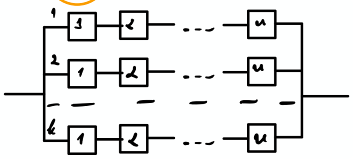
- **Pentru fiabilitatea sistemelor nedecompozabile**

  → se va lua pe cazuri < modulul "problemă" funcționează / modulul "problemă" nu funcționează și

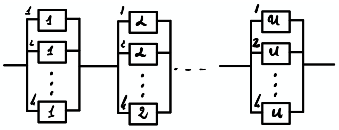
  se va face cu "sau"

- **Rezervare bimodală**
  - $R_{SP} = 1 - (1 - R^n)^k$
  - $R_{PS} = \left[1 - (1 - R)^k\right]^n$

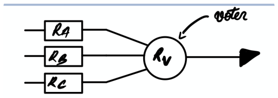
SP



PS

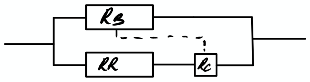


## Structuri cu votare majoritară



- $$R_{RVM} = \left[ \sum_{i=0}^{n-1} C_{2n-1}^{i} R^{2n-1-i} (1-R)^{i} \right] R_V$$

## Structuri cu un element de rezervă



- $$R_S(t) = \frac{\lambda_R}{\lambda_R - \lambda_B} \cdot R_B(t) - \frac{\lambda_B}{\lambda_R - \lambda_B} R_R(t)$$
- $MTBF_S = MTBF_B + MTBF_R$

### → cu k-1 elemente de rezervă:

- $$R_{S,k} = \sum_{j=1}^{k} \left[ \prod_{\substack{i=1 \\ i \neq j}}^{k} \frac{\lambda_i}{\lambda_i - \lambda_j} e^{-\lambda_i t} \right]$$
- $$MTBF_S = \sum_{i=1}^{k} MTBF_i$$

### → module identice:

- $$R_{S,k} = \sum_{j=1}^{k} \frac{(\lambda t)^{j-1}}{(j-1)!} e^{-\lambda t}$$
- $MTBF = K \cdot MTBF$

## Structuri cu backup activ

### → module identice:

- $$R_S(t) = 1 + \frac{\lambda}{\lambda_u} ( 1 - e^{-\lambda_u t} ) ] e^{-\lambda t}$$
- $$MTBF = \frac{1}{\lambda} + \frac{1}{\lambda + \lambda_u}$$

- Lanțuri Markov - la timpi discreți
  - $$A = \begin{bmatrix} p_{11} & p_{12} & \cdots & p_{1n} \\ \sim & - & - & - \\ p_{n1} & p_{n2} & \cdots & p_{nn} \end{bmatrix}$$ "Matrice de probabilități"
  - $\mathbb{P}_1(t) \circ A = \mathbb{P}_1(t+\Delta t)$

- Steady State
  - $[q_1 \ q_2] \circ A = [q_1 \ q_2]$

- Procese Markov - timp de observație continuu
  - $\lambda_{ij}$ = "densitate probabilitate de tranziție"

    $$\lambda_{ij} = \lim_{\Delta t \to 0} \frac{P\{X(t+\Delta t) = S_j \mid X(t) = S_i\}}{\Delta t}$$
  - $$A^* = \begin{bmatrix} -\sum_{i=2}^{n} \lambda_{1i} & \lambda_{12} & \lambda_{13} & --- & \lambda_{1n} \\ \sim & - & - & - & - \\ \lambda_{n1} & \lambda_{n2} & - & - \, --- & -\sum_{i=1}^{n} \lambda_{in} \end{bmatrix}$$
  - $P'(t) = P(t) \cdot A^*$